

Hosting a Private Helm Repository using Apache Web Server

Helm is a package manager for Kubernetes, and uses a package format called Chart. A Chart consists of several Kubernetes YAML files. Helm packages a set of Charts that describe Kubernetes resources. Let's learn how to create our own private Helm repository to host Helm Charts using Apache Web Server.



Kubernetes is a portable, extensible and open source platform for managing containerised workloads and services. It facilitates both declarative configuration and automation, and provides a way to run distributed systems resiliently. K8s allows you to take care of self-healing, scaling, load balancing and failover of your applications.

In the last article on Helm, we discussed its usage, creating your own Helm Chart and installation methods. Now let's create our own private Helm repository to host Helm Charts using Apache Web Server with a self-signed SSL certificate.

The requirements are:

- Kubernetes
- Helm
- Kubectl
- Apache Web Server

Install and configure Apache Web Server

Install Apache Web Server with a self-signed SSL certificate, with basic authentication on CentOS 7.

Let's update and install Apache:

```
# yum update
# yum install httpd mod_ssl -y
```

Now create a self-signed SSL certificate:

```
# mkdir /etc/httpd/ssl
# openssl req -x509 -nodes -days 365
-newkey rsa:2048 -keyout /etc/httpd/
ssl/helm_private.key -out /etc/httpd/
ssl/helm_repo.crt
```

After answering the questions, you will get `/etc/httpd/ssl/helm_private.key` and `/etc/httpd/ssl/helm_repo.crt`.

Now, configure this SSL certificate in `httpd` configuration.

Open `/etc/httpd/conf.d/ssl.conf` and edit the following lines:

```
SSLCertificateFile /etc/httpd/ssl/helm_
private.key
SSLCertificateKeyFile /etc/httpd/ssl/
helm_repo.crt
```

